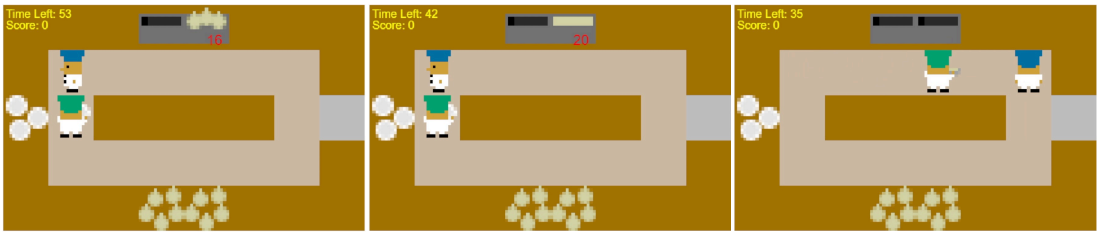
#### 7.1.1 Experimental Setting

We divided each training batch into four parts, the ratio indicating the proportion of data generated by self-play and data generated by playing with strategies from the cooperative incompatible distribution. We omitted the 4:0 ratio as it would result in the framework degenerating into self-play.

#### 7.1.2 Results

Fig. 7 shows the mean rewards of episodes over 400 time steps of gameplay when paired with the unseen human proxy partner $H_{proxy}$ (Carroll et al., 2020). We found that $\text{COLE}_{\text{SV}}$ with ratios 0:4 and 1:3 achieved better performance than the other ratios. In particular, $\text{COLE}_{\text{SV}}$, with a ratio of 1:3, outperformed the other methods in the Cramped Room, Coordination Ring, and Counter Circuit layouts. On the Forced Coordination layout, which is particularly challenging for cooperation due to the separated regions, all four ratios performed similarly on average across different starting positions. Interestingly, $\text{COLE}_{\text{SV}}$ with only the cooperative compatible objective (ratio 0:4) performed better on the Asymmetric Advantages and Forced Coordination layouts when paired with the human proxy partner. Effectiveness evaluations indicate that the combination of individual and cooperatively compatible objectives is crucial to improving performance with unseen partners. In general, we choose the ratio of 1:3 as the best choice.

We further visualize the trajectories produced by $\text{COLE}_{\text{SV}}$ 1:3 and 0:4 with human proxy and expert partners in Overcooked on our demo page. Fig. 8 presents three screenshots of the $\text{COLE}_{\text{SV}}$ 0:4 model (blue player) that collaborates with one of the expert partners, the PBT model (green player). The case illustrates the importance of the individual objects in zero-shot coordination with expert partners. Frame A is a screenshot taken at 53s when the two players start to impede each other. The PBT model has taken the plate and wants to load and serve the dish. The blue player wants to take the plate but does not know how to change the objective to allow the green player to load the dish. After blocking for about 11s, the blue player starts to move and lets the green player go to the pots (Frame B). However, the process is not smooth and takes 7s to reach Frame C. This phenomenon does not occur in $\text{COLE}_{\text{SV}}$ 1:3 coordination with expert partners, which shows that including individual objectives might improve the cooperative ability with expert partners.




**Frame A : time left 53s** **Frame B : time left 42s** **Frame C : time left 35s**

Figure 8: Trajectory snapshots of the $\text{COLE}_{\text{SV}}$ 0:4 model (blue) with one of the expert partners - PBT model (green).

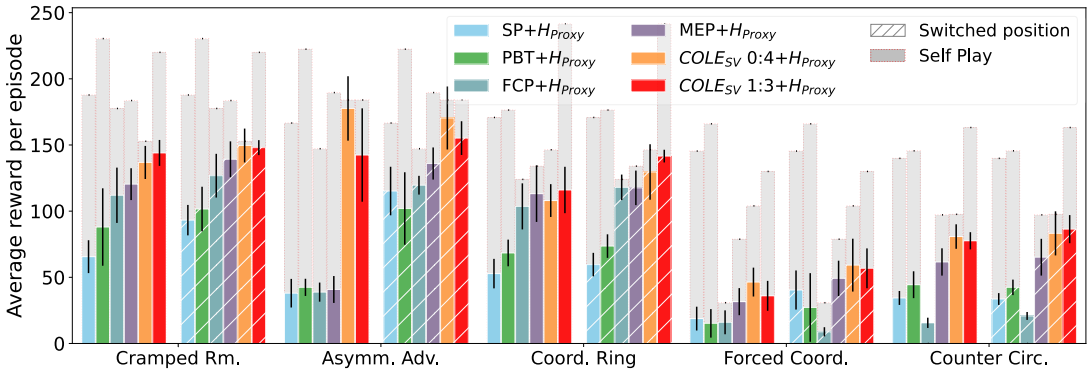



Figure 9: Performance with the human proxy partner. The performance of $\text{COLE}_{\text{SV}}$ with human proxy partners is presented in terms of mean episode rewards over 400 timesteps trajectories for different objective ratios of 0:4 and 1:3, when paired with the unseen human proxy partner $H_{proxy}$. The ratios 0:4 and 1:3 denote varying proportions between individual and cooperative compatible objectives. The results include the mean and standard error over five different random seeds. The gray bars indicate the rewards obtained when playing with themselves; the hashed bars indicate the performance when starting positions are switched.

### 7.2 Evaluation with Human Proxy and AI Agents

To thoroughly assess the ZSC ability, we evaluated the algorithms with unseen human proxy and expert partners. We compare our method with other methods, including self-play (Tesauro, 1994; Carroll et al., 2020), PBT (Jaderberg et al., 2017; Carroll et al., 2020), FCP (Strouse et al., 2021), and MEP (Zhao et al., 2021), all of which use PPO (Schulman, Wolski, Dhariwal, Radford, & Klimov, 2017) as the RL algorithm. We use the human proxy model $H_{proxy}$ proposed in (Carroll et al., 2020) as human proxy partners and the models trained with baselines and $\text{COLE}_{\text{SV}}$ as expert partners.

#### 7.2.1 Experimental Setting

We adopted two sets of evaluation protocols for the evaluation. The first protocol involves playing with a trained human model $H_{proxy}$ trained in behavior cloning. Due to the quality and quantity of human data used for behavior cloning to train the human model is limited, the capabilities of the human proxy models are limited. Therefore, we use an additional evaluation protocol to coordinate with unseen expert partners. We selected the best models of our reproduced baselines and $\text{COLE}_{\text{SV}}$ 0:4 and 1:3 as expert partners. The mean of the rewards is recorded as the performance of each method in collaborating with expert teammates. Appendix C and Appendix D give details of the implementation of $\text{COLE}_{\text{SV}}$ and baselines.

Table 2: Performance with expert partners. Mean episode rewards over 1 min trajectories for baselines and $\text{COLE}_{\text{SV}}$ with ratio 0:4, 1:3. The ratios 0:4 and 1:3 denote varying proportions between individual and cooperative compatible objectives. Each column represents a different expert group, in which the result is the mean reward for each model playing with all others.

| Layout | Ratio | Baselines | | | | COLEs |
|---|---|---|---|---|---|---|
| | | SP | PBT | FCP | MEP | |
| Cramped Rm. | 0:4 | 153.00 | 198.50 | 199.83 | 178.83 | 169.76 |
| | 1:3 | 165.67 | 209.83 | 207.17 | 196.83 | **212.80** |
| Asymm.Adv. | 0:4 | 108.17 | 164.83 | 175.50 | 179.83 | **182.80** |
| | 1:3 | 108.17 | 161.50 | 172.17 | 179.83 | 178.80 |
| Coord. Ring | 0:4 | 132.00 | 106.83 | 142.67 | 130.67 | 118.08 |
| | 1:3 | 133.33 | 158.83 | 144.00 | 124.67 | **166.32** |
| Forced Coord. | 0:4 | 58.33 | 61.33 | 50.50 | 79.33 | 46.40 |
| | 1:3 | 61.50 | 70.33 | 62.33 | 38.00 | **86.40** |
| Counter Circ. | 0:4 | 44.17 | 48.33 | 60.33 | 21.33 | 90.72 |
| | 1:3 | 65.67 | 64.00 | 46.50 | 76.67 | **105.84** |

#### 7.2.2 Results with Human Proxy and AI Agents

Fig. 9 presents the performance of SP, PBT, MEP, and $\text{COLE}_{\text{SV}}$ with 0:4 and 1:3 when cooperating with human proxy partners. We observed that different starting positions on the left and right in asymmetric layouts resulted in significant performance differences for the baselines. For example, in the Asymmetric Advantages, the cumulative rewards of all baselines in the left position were nearly one-third of those in the right position. On the contrary, $\text{COLE}_{\text{SV}}$ performed well at the left and right positions.

As shown in Fig. 9, $\text{COLE}_{\text{SV}}$ outperforms other methods in all five layouts when paired with the human proxy model. Interestingly, $\text{COLE}_{\text{SV}}$ 0:4 with only the cooperatively compatible objective achieves better performance than $\text{COLE}_{\text{SV}}$ 1:3 on some layouts, such as Asymmetric Advantages. However, the self-play rewards of $\text{COLE}_{\text{SV}}$ 0:4 are much lower than $\text{COLE}_{\text{SV}}$ 1:3 and even other baselines. The objective function of $\text{COLE}_{\text{SV}}$ 0:4 consists of only cooperative compatible term $\mathbb{E}_{p\sim\phi}\mathbf{w}(s_n, p)$. We believe that focusing solely on cooperative compatible objectives may lead to learning stagnation when collaborating with low-performing partners sampled from distribution $\phi$. Consequently, the self-play rewards may be limited. Furthermore, the performance with unseen experts of $\text{COLE}_{\text{SV}}$ 0:4 as shown in Table 2, is sometimes lower than the baselines.

Table 2 presents the outcomes of each method when cooperating with expert partners. Each column in the table represents different expert groups, including four baselines and